

Running now

CIFAR-10 Classifier

Run skip-v2, train mode

Epoch 9 of 20

loss 0.71 val/acc 73.8%

Open run

Training

Speech Commands

Run baseline, train mode

The runner notebook is ready in your Drive. Press Run all in Colab and this run will connect by itself.

Open in Colab

Waiting for Colab

Recent projects

All projects

CIFAR-10 Classifier

Image

2 runs, best val/acc 74.1%

ViT Flowers

Image

Last run failed at epoch 1

Speech Commands

Audio

First run waiting for Colab

Recently edited blocks

Open library

Conv Block	Layer	[B,C,H,W] to [B,C,H,W]	v3, today
Speech Commands	Data	waveform, label	v1, today
Focal Loss	Loss	pred, target	v2, yesterday

Projects

Each project is one trainer. Its runs and files are saved inside it.

+ New project

Modality All

Name	Modality	Last run	Status	Best result	Updated
CIFAR-10 Classifier	Image	skip-v2	● Training	val/acc 74.1%	2 min ago
Speech Commands	Audio	baseline	● Waiting for Colab	None yet	Today
ViT Flowers	Image	baseline	● Failed	None yet	Yesterday
Sentiment Transformer	Text	None	● Not run yet	None yet	3 days ago

Start a new project

Give it a name and an empty trainer canvas opens. Start with a data block and end with a loss block.

Project name

Create



CIFAR-10 Classifier

Saved 1 min ago

Build

Runs 2

✓ Valid, 10 connections



Config

Run



START

 **CIFAR-10**
train split, 32 px

image

label

 **Conv Block**
3 to 32, kernel 3

x

out

 **Conv Block**
32 to 64, 2 outputs

x

main

skip

 **Conv Block**
64 to 64, kernel 3

x

out

 **Add**
2 inputs, same shape

a

b

out

 **MaxPool**
kernel 2

x

out

 **Flatten**
to [B, 16384]

x

out

 **Linear**
16384 to 10

x

out

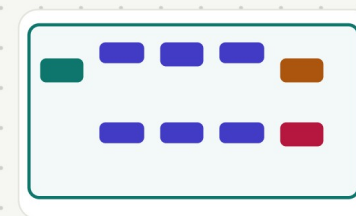
 **Cross Entropy**
mean, ends the step

pred

target

END, backward and optimizer step

100% Fit Tidy layout



skip-v2

● Training

Open in Colab

■ Stop run

✓ Queued
14:01

✓ Waiting for Colab
Run all pressed 14:02

✓ Connected
Bundle downloaded 14:03

● Training
Epoch 9 of 20

Finished
Not yet

Last heartbeat **12 s ago** Mode **Train** Elapsed **11 min** Runtime **Colab, T4 GPU** Config **lr 0.0005, 20 epochs, batch 64**

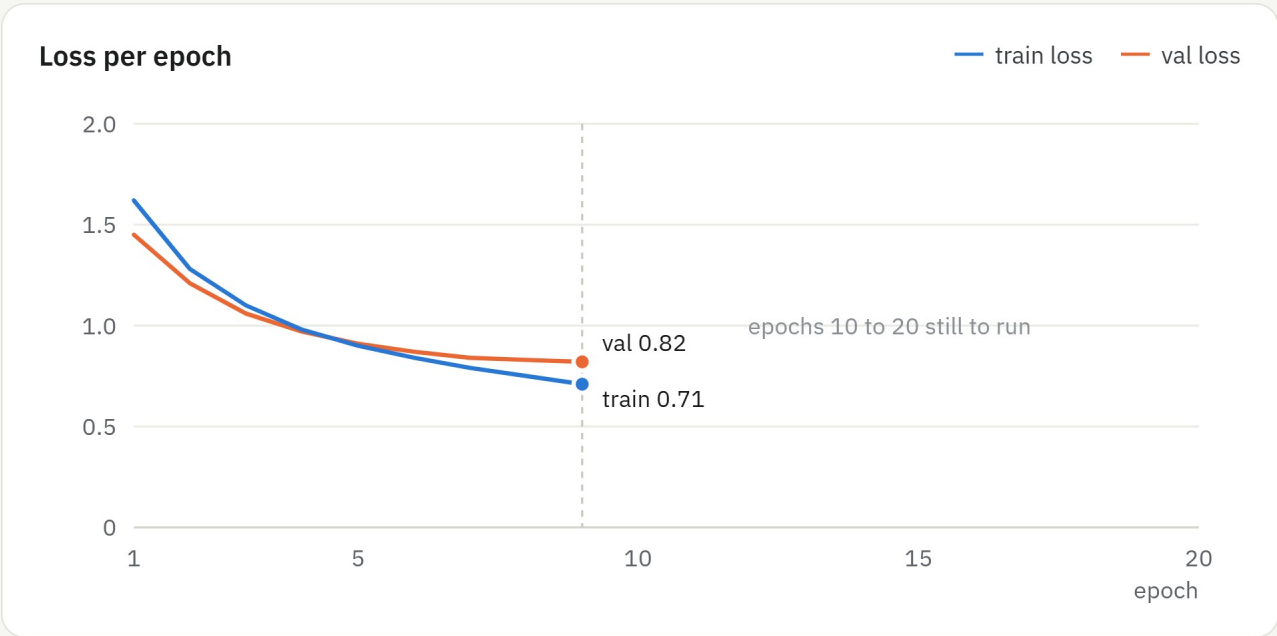
Overview Charts Plots (2) Logs Files (6)

Epoch
9 of 20

Train loss
0.71

Val loss
0.82

Best val/acc
74.1% epoch 8



● Live log

Level All ▼

14:12:44 INFO step=3500 loss=0.6981

14:13:02 INFO step=3550 loss=0.7012

14:13:20 INFO step=3600 loss=0.7134

14:13:20 INFO step=3600 lr/optimizer=5.000000e-04

14:13:38 INFO Epoch 8 complete, avg(loss=0.7498)

14:13:38 INFO Epoch 8 time=78.41s total_elapsed=631.20s

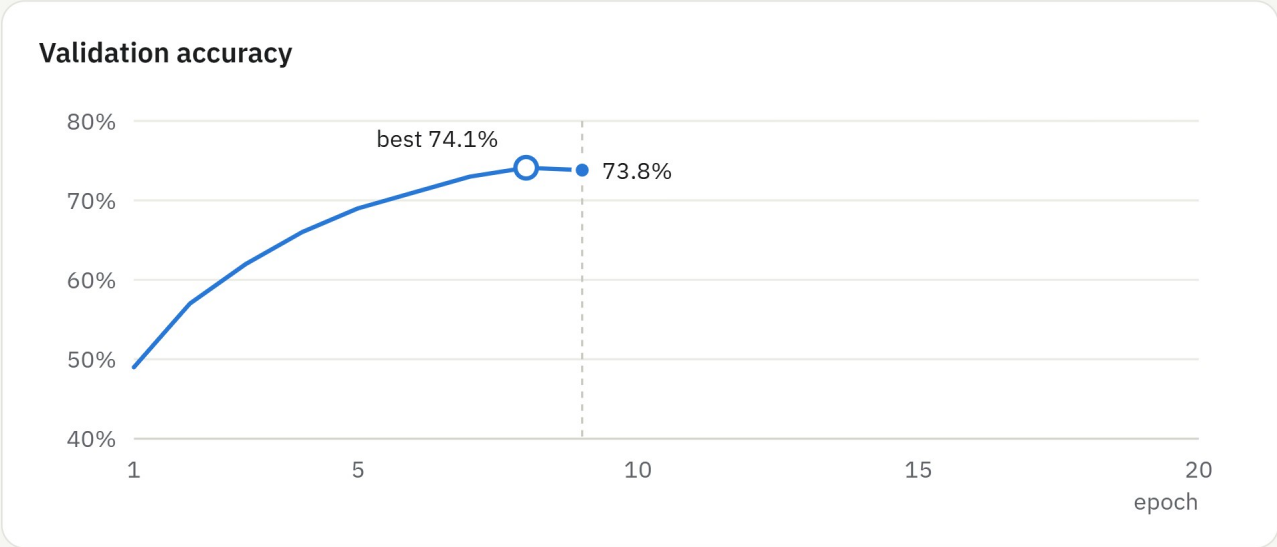
14:13:39 INFO Saved checkpoint to runs/skip-v2/latest.pt

14:13:41 INFO Uploaded latest.pt to the project

14:13:59 INFO step=3650 loss=0.7208

14:14:17 INFO step=3700 loss=0.7066

Open full log



Files

Saved in this project

latest.pt2.7 MB

epoch_5.pt2.7 MB

train_curves_epoch5.png64 KB

metrics.csv92 KB

config.yaml1 KB

blocks/ (code used)6 KB

Download all (.zip)

Resume from latest



CIFAR-10 Classifier

Saved 1 min ago

Build

Runs 2

Valid, 10 connections



Config

Run



Add blocks

Click to add, or drag onto the canvas

Search blocks

All

Image

Text

Audio



Data

2



CIFAR-10

Built-in images. Outputs image and label.

Speech Commands

Hugging Face audio. Outputs waveform and label.

+ Add a dataset



Layer

7



Conv Block

Convolution, batch norm and ReLU.

Residual Block

Two convs. Outputs main and skip.

MaxPool

Halves height and width.

Linear

Fully connected layer.

Show 3 more: Flatten, Patch Embedding, Transformer Encoder



Merge

2



Loss

2



Callback

1



Missing something?

Create a new block

Conv Block

3 to 32, kernel 3

out



Conv Block

32 to 64, 2 outputs

x

main

skip



Conv Block

64 to 64, kernel 3

out



Add

2 inputs, same shape

a

b

out

skip

MaxPool

kernel 2

out



Flatten

to [B, 16384]

x

out



Linear

16384 to 10

x

out



Cross Entropy

mean, ends the step

pred

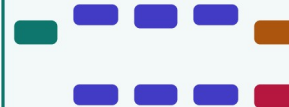
target

END, backward and optimizer step

label



Drop Residual Block anywhere on the canvas, then connect its dots





CIFAR-10 Classifier

Saved 1 min ago

Build

Runs 2

Valid, 10 connections



Config

Run



START

CIFAR-10
train split, 32 px

image

label

Conv Block
3 to 32, kernel 3

x

out

Conv Block
32 to 64, 2 outputs

x

main

skip

MaxPool
kernel 2

x

out

Flatten
to [B, 16384]

x

out

label

Conv Block

Layer, v3

Parameters

in_channels	out_channels	kernel_size
32	64	3
in [B,32,32,32]		out [B,64,32,32]

Outputs

main	to Conv Block (64 to 64)
skip	to Add, input b (skip)

Sending an output to a Merge block further down makes a skip connection. Shapes are checked where they meet.

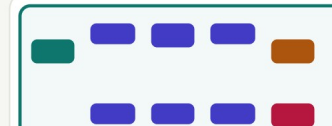
☐ **Freeze this block**
Its weights are not trained, for example a JEPA target encoder.

Edit code

Duplicate

Delete

100% Fit Tidy layout



Project config

×

CIFAR-10 Classifier. Used for every run of this project.

Model

9 blocks on the canvas

Training

20 epochs, Adam

Data

batch 64, test as validation

Saving

checkpoints and logging

Plots

every 5 epochs, 3 metrics

Early stopping

on, val/loss, patience 3

Run length

Epochs

20

Seed

42

Device

GPU

Optimizer

Type

Adam

Learning rate

0.0005

Weight decay

0

Updates every trainable block on the canvas. Frozen blocks are left out.

Loss

Cross Entropy

set on the canvas, the last block

config.yaml, live preview

Copy

```
epochs: 20
seed: 42
data:
  name: cifar10
  batch_size: 64
  num_workers: 2
model:
  name: cifar10_classifier
  optimizers:
    optimizer:
      type: Adam
      lr: 0.0005 # changed
      weight_decay: 0
  criteria:
    main:
      type: cross_entropy
callbacks:
  checkpoint:
    save_every_steps: 1000
    keep_every_epochs: 5
  logging:
    log_every_steps: 50
  plot:
    plot_every_epochs: 5
early_stopping:
  monitor: val/loss
  patience: 3
```

1 change not saved

Cancel

Save config

Blocks

Your library. Datasets are blocks too, and every block can be used in any project.

DATA

Built-in

CIFAR-10

out image [3,32,32]

out label

Image, used in 1 project

DATA

Hugging Face

Speech Commands

out waveform [16000]

out label

Audio, used in 1 project

LAYER

v3

Conv Block

in [B,C,H,W]

out [B,C,H,W]

Image, used in 2 projects

LAYER

v1

Residual Block

in [B,C,H,W]

out main, skip

Image, 2 parallel outputs

LAYER

v2

Patch Embedding

in [B,3,H,W]

out [B,N,D]

Image, used in 1 project

LAYER

Built-in

Transformer Encoder

in [B,N,D]

out [B,N,D]

Text and image

LAYER

v1

Spectrogram

in [B,T]

out [B,F,T]

Audio, used in 1 project

MERGE

Built-in

Add

in a, b (same shape)

out a + b

Any modality

MERGE

Built-in

Concat

in 2 or more

out joined on a dim

Any modality

LOSS

Built-in

Cross Entropy

in pred, target

ends the canvas

Classification

LOSS

v2

Focal Loss

in pred, target

ends the canvas

Custom, uploaded file

CALLBACK

v1

Early Stop on Plateau

runs on epoch end

no connections

Any project

Drop a .py file here to turn its class into a block, or drop data files to add a dataset

Conv Block

Layer

Image

Used in 2 projects

Write code

⤴ Upload .py file

Describe with AI

Later

conv_block.py

Drop a .py file here to replace it

```
1 import torch.nn as nn
2
3
4 class ConvBlock(nn.Module):
5     def __init__(self, in_channels=3, out_channels=32, kernel_size=3):
6         super().__init__()
7         self.conv = nn.Conv2d(in_channels, out_channels,
8                               kernel_size, padding=kernel_size // 2)
9         self.bn = nn.BatchNorm2d(out_channels)
10        self.act = nn.ReLU()
11
12    def forward(self, x):
13        return self.act(self.bn(self.conv(x)))
```

✓ Validate passed

Ran once on dummy data: input x [4, 3, 32, 32] gave out [4, 32, 32, 32].

Found an nn.Module with forward(x) and 3 parameters in __init__.

Name

Conv Block

Registry name

conv_block

Type

Layer ▾

Modality

Image ▾

Parameters

Read from __init__, shown on the card

Name	Type	Default
in_channels	int	3
out_channels	int	32
kernel_size	int	3

Inputs

○ x tensor [B, C, H, W]

Outputs

1 output

○ out tensor [B, C, H, W]

+ Add parallel output

A second output, for example skip, gets its own dot on the card and can go straight to an Add or Concat block.

CIFAR-10

■ Data

Image

Built-in, 10 classes

Validate

Use in a project

Where the data comes from

☒ Built-in

☐ Upload files

☐ Kaggle

☐ Google Drive

☐ Hugging Face

☐ URL

Sample preview

Show other samples



label 3, cat



label 8, ship



label 6, frog



label 9, truck



label 2, bird



label 0, airplane



label 7, horse



label 4, deer

Images here are placeholders. The real app shows the 32 x 32 samples. Audio data shows a waveform with a play button, and text shows the tokens.

Fields

Each field becomes an output on the card

Field	Type	Shape	On the card
image	float32	[3, 32, 32]	output
label	int64	[] values 0 to 9	output

Splits

train

50,000

test

10,000

Block parameters

Image size

Split

32

train

☒ Download on Colab if missing

Batch size and workers are set per project in the Config menu.

How it looks on the canvas

■ CIFAR-10

DATA

image

label